

A Passive Battery Management System for Fast Balancing of Four LiFePO₄ Cells

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Abstract—In this paper, a Battery Management System (BMS) is designed and implemented to enable fast balancing during charging of four Lithium Iron Phosphate (LiFePO₄) cells connected in series, designated to replace the classical Lead-Acid battery of a vehicle. The BMS detects when the battery pack is charged and it enables passive balancing of charged cells, which are bypassed through high power resistors. The measured parameters (voltages, current, and temperature) and charging/discharging states can be displayed on LCD and transmitted to other Electronic Control Units (ECUs) using Controller Area Networks (CAN).

Keywords—Battery Management Systems; passive balancing; LiFePO₄ cells

I. INTRODUCTION

In our days, battery technology is growing very fast and provides practical and efficient solutions to store energy in traction and power applications like Electric Vehicles (EV), Hybrid Electric Vehicles (HEV), Plug-in Hybrid Electric Vehicles (PHEV), industrial vehicles, marine, Energy Storage Systems (ESS), and Uninterruptible Power Supplies (UPS) [1].

The lithium batteries are used for energy storage because of its high power density and long lifespan, increasing the driving range of a vehicle. Depending on the application, a large number of cells are connected in series to increase the voltage or in parallel to increase the current of the battery pack. Each battery when operating in series has a slightly different capacity due to manufacturing tolerances and environmental conditions. After several charge/discharge cycles, an individual cell can have lower performance and capacity, so the capacity of the entire battery pack will be reduced [1], [2].

A Battery Management System (BMS) has the main role to make the use of battery as safe, reliable, and cost efficient. The most important feature of a BMS is cell balancing, which equally distributes the energy among battery cells. Also, the BMS should contain algorithms to precisely measure and estimate the State of Charge (SOC), Depth of Discharge (DOD), State of Health (SOH), Remaining Useful Life (RUL) and to offer protection for overcharging, deep discharging and temperature protection, thus maintaining each cell of the battery pack within its safe operating range [3]-[5]. The voltage difference among cells connected in series can be caused by a SOC imbalance, a total capacity imbalance or an internal impedance imbalance [6].

Generally, there are two cell balancing methods: passive and active. The main difference between them is that for passive balancing the extra energy of the most charged cells is dissipated through switched resistors and for active balancing the energy from the strong cells is transferred to the weak cells [6].

Passive cell balancing is the most widely used due to its simplicity, reliability and low-cost characteristics. A classical BMS assures small balancing currents, up to hundreds of mA, the balancing of a high capacity battery pack being completely realized in many charging/discharging cycles [7]-[8].

In this paper, we propose a passive BMS, which can assure high balancing currents, for a fast balancing of the cells. In Section II, the implementation of the proposed BMS is described, and in Section III the measurement results are presented. Finally, Section IV concludes this paper.

II. THE IMPLEMENTED BATTERY MANAGEMENT SYSTEM

The implemented BMS is designed for a pack of four Lithium Iron Phosphate (LiFePO₄) cells connected in series (1P4S), with a pack nominal voltage of 12.8 V, which can replace the classical Lead-Acid battery in a vehicle or 12 V applications. The cells are CA180FA from CALB (China Aviation Lithium Battery Limited) and have following specifications: nominal voltage 3.20 V, nominal capacity 180 Ah, operating voltage 2.60 - 3.60 V, discharging minimum cut-off voltage 2.50 V, charging maximum cut-off voltage 3.65 V, recommended charging / discharging current 0.3C (56 A), maximum short-time discharging current 10C (1800 A), 2000 life cycles at 80% DOD, internal impedance $\leq 0.6 \text{ m}\Omega$ at 1 kHz AC, dimensions 278x180x71 mm, weight 5.8 kg [9].

The standard charging method for these LiFePO₄ cells is constant current (CC) charging at 0.3C until the voltage is bigger than 3.60 V (95% SOC), then constant voltage (CV) charging at 3.60 V until the current becomes less than 0.05C (100% SOC). A rapid charging can be realized at 1.0C [9].

A. The hardware implementation

The proposed passive BMS adopts a centralized architecture (wires for each cell), static balancing (during only the charging state), passive balancing (the surplus of energy for one cell is consumed through power resistors), parallel balancing (more cells can be balanced at a time), high balancing currents up to 7 A, high precision (the voltage difference between cells is less than 25 mV) [10].

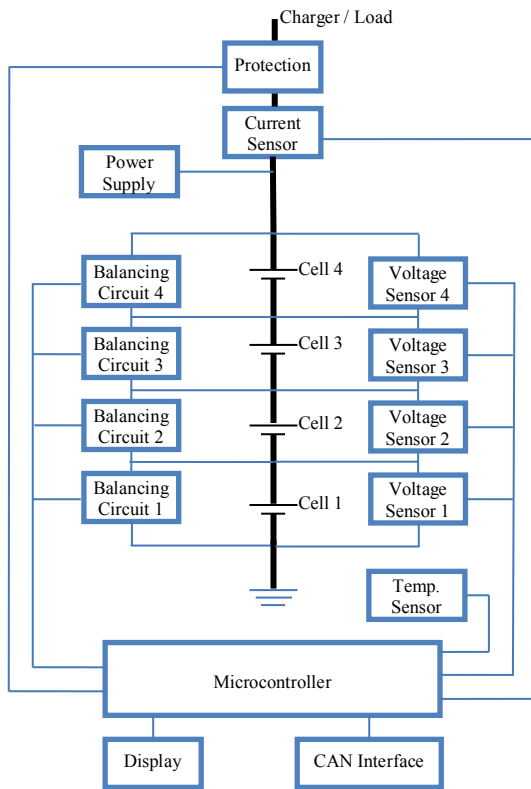


Fig. 1. The block architecture of the proposed passive BMS.

Other main features of the proposed BMS include: voltage monitoring of individual cells, voltage, current and temperature monitoring of the battery pack, overcharging and deep discharging monitoring and protection, internal error fault detection, transmission of the measured parameters and states through the CAN network.

The BMS architecture is shown in Fig. 1. The four battery cells are shown in the center. The acquisition blocks for current, voltage and temperature are shown on the right side. The acquired data are processed by the ATMEGA328P microcontroller and printed on display or transmitted through the CAN interface. Based on the proposed balancing algorithm, the microcontroller controls the balancing circuits of the four cells and the protection circuit.

The balancing circuit for each cell is composed of two identical power transistor-resistor branches (see Fig. 2), each branch having the role of steering part of the charging current such that it bypasses the correspondent battery cell. The power resistors R2, R3 have very low values (0.68 Ω) and the PNP Darlington transistors Q1, Q2 (BDX54C) are able to withstand large current values such that a shorter balancing time is achieved. For a cell voltage of 3.33 V, the collector-emitter voltage of Q1 is set to 1.11 V and the voltage drop on R2 is 2.22 V, the current being 3.25A, and the total balancing current being 6.5 A. To control the power transistors we use a driver circuit formed by an optocoupler U1 (PC817) and resistors R1, R4 and R5, which have bias purpose. Basically, a control signal CTRL1 coming from the microcontroller turns the optocoupler on and off, which, in turn, turns the balancing transistors on and off.

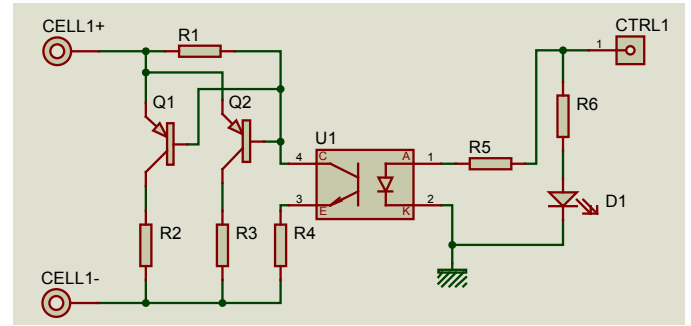


Fig. 2. The balancing circuit for Cell 1.

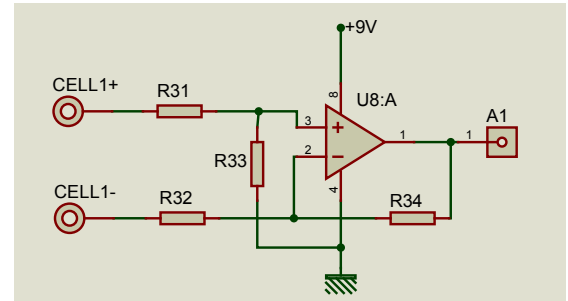


Fig. 3. The voltage measuring circuit for Cell 1.

The voltage of each cell is measured by a classical operational amplifier LM358 used as a differential amplifier (see Fig. 3). In this way, the electronic circuit for each cell remains the same. The output A1 is connected to an analog input of the microcontroller. Because the voltage of each cell is less than 5 V, the amplification factor can be unitary, so all resistors have the same value (100 k Ω).

The current of the battery pack is measured in both charging and discharging cycles using a LEM HTFS-400P Hall current sensor, having the rms nominal current of 400 A.

The temperature of the battery pack is measured using a classical LM35DT temperature sensor.

The measured parameters (voltages of the 4 cells, the current and the temperature of the battery pack) are displayed on a 20x4 characters Liquid Cristal Display (LCD).

The measured parameters and the states of the BMS are transmitted through the CAN interface implemented with the MCP2515 controller and the TJA1050 transceiver.

The power supply provides the necessary voltages (5 V and 9 V) to power the microcontroller, the sensors, the LCD and the CAN interface. It is based on LM7805 and LM7809 voltage regulators along with the needed filtering components. The input voltage is assured by the battery pack itself.

The protection circuit contains a power relay driven by a low power transistor, which disconnects the charger or the load when the minimum or maximum cut-off voltages for at least one cell are detected.

The implemented passive BMS is shown in Fig. 4. The control board is shown on the right side and includes the microcontroller, the LCD display, the protection circuit, and the CAN interface. The power board is on the left side and



Fig. 4. The implemented passive BMS.

includes the measuring circuits for the cells voltages, current and temperature of battery pack, the circuits for cells balancing with power transistors and resistors, and the power supply.

B. The software implementation

The software part of this BMS consists of the code loaded onto the microcontroller. The main steps of the balancing algorithm are as follows:

- the battery pack parameters are measured and displayed on LCD: voltages for each cell, pack voltage, pack current and pack temperature. To improve the accuracy of measurements, averaging is applied over 50 samples, the voltages being displayed with an accuracy of 1 mV and the current with 10 mA;
- based on the sign of the current, the algorithm checks if the battery pack is charging or discharging and activates the cells balancing or not;
- if the battery pack is charging and the difference between the cells voltages exceeds a certain threshold, defined as the balancing voltage which is set to 25 mV, the balancing circuits for the most charged cells are activated. If the voltage of a cell is greater than 3.60 V or the pack temperature is greater than 45°C, then the protection relay is activated and the charging process is interrupted;
- if the battery pack is discharging and the voltage of a cell is smaller than 2.55 V, then the protection relay is activated and the discharging process is interrupted.

III. MEASUREMENT RESULTS

The testing platform for the implemented BMS contains the four LiFePO₄ cells, the BMS itself, an analog power supply to charge the battery pack and different loads to discharge it, see Fig. 4. The testing platform can be used to analyze different balancing algorithms based on cell voltage, state of charge or cell capacity. The balancing efficiency can be measured for different scenarios, when one or more cells are discharged.

At the beginning of each discharging/charging cycle, the battery pack must be charged at 50% SOC. To equally balance the cells at 3.33 V, the four cells are connected in parallel for 48 hours [11].

To discharge certain cells we use three halogen lamps of 12 V / 55 W and one resistive load of 4 Ω / 45 W, see Fig. 4. If the loads are manually connected to one desired cell, the measured discharging current is 7.35 A. During discharging, the BMS monitors the cells and if the voltage of one cell drops below 2.55 V, the discharging process is stopped to protect that cell from damage.



Fig. 5. The testing platform during the discharging cycle.

To charge the entire battery pack, an adjustable power supply is used, the maximum charging current being 6 A. Also, the power supply has controlling options for constant current (CC) and for constant voltage (CV). The charging process evolves in two stages: firstly we have CC charging at 6 A until the voltage of the battery pack reaches 3.60 V x 4 cells = 14.4 V and then we have CV charging until the current decreases significantly.

For the first test, the cells are discharged at different rates, see Fig. 6. At the beginning of charging, the cells voltages are 3.29, 3.29, 3.15 and 3.00 V, which corresponds to 39%, 39%, 8% and 4% SOC, the SOC of the battery pack being 22.5%, and the SOC imbalance 66% [5], [11]. The cells voltages measured during charging are shown in Fig. 6, the measurements being taken every one hour.

The difference between cells becomes lower than the balancing voltage of 25 mV after the balancing time of 18 hours, and remains constant until the charging is stopped after 27 hours. For a balancing current of 6.5 A and a cell capacity of 180 Ah, results that a 3.6% balancing can occur in one hour, and the battery pack can be completely balanced in 18 hours.

Also, from Fig. 6 we can observe that the charged cells 1 and 2 are easily discharged through the balancing resistors and transistors during the first 18 hours because the balancing current of 6.5 A is bigger than the charging current of 6 A.

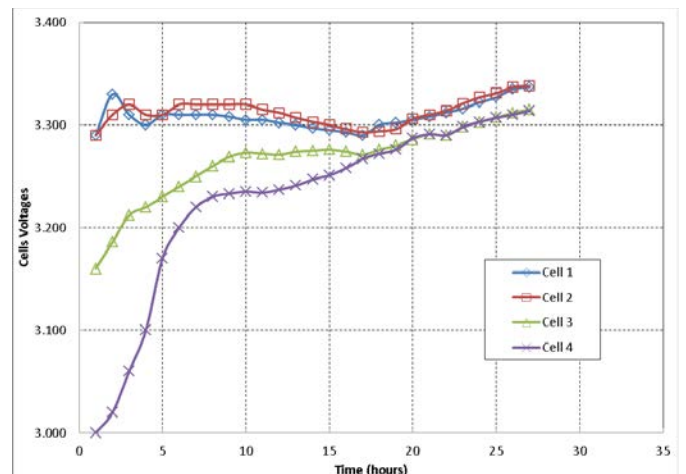


Fig. 6. The cells voltages during charging – Test 1.

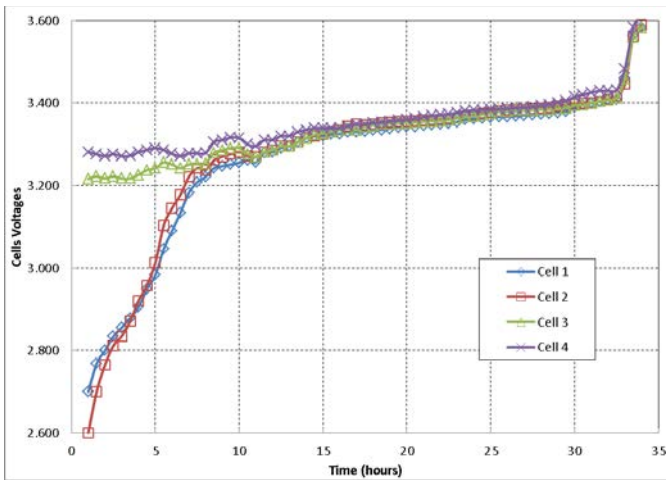


Fig. 7. The cells voltages during charging – Test 2.

For the second test, see Fig. 7, the cells 1 and 2 are fully discharged, and the cells 3 and 4 are partially discharged. The cells voltages at the beginning of charging are 2.70, 2.60, 3.21 and 3.28 V (2%, 1%, 12% and 23% SOC), the SOC of the battery pack being 9.5%, and the SOC imbalance 54%. Because a 3.6% balancing can occur in one hour, the balancing time results to be 15 hours. This value can be derived also from the experimental measurements from Fig. 7.

Because for the cells 1 and 2 we have made a complete discharge/charge cycle, we can estimate their measured capacity as being the charging current multiplied by the charging time, $6\text{ A} \cdot 34\text{ hours} = 204\text{ Ah}$. The measured capacity is a little bigger than the nominal capacity of 180 Ah or the capacity rated by the manufacturer of 200 Ah.

During the balancing of one cell, the measured balancing current is 6.5 A, and the dissipated power is approximately 7.25 W on each resistor and 3.6 W on each transistor. In the worst case scenario of three balancing cells, the total dissipated power by the BMS can reach up to 65 W.

To decrease the dissipated energy during balancing, the power transistors are switched on for 5 seconds and off for 3 seconds. Also, to increase the power handled by the power resistors, we mounted additional radiators on them.

Analyzing the measurements results after more charging and discharging cycles, to minimize the case of a damaged cell because of battery hysteresis [11], we propose to use the following limiting values of 3.0 V for discharging minimum cut-off voltage and 3.55 V for charging maximum cut-off voltage instead of the limits (2.50 V and 3.65 V) specified by the cells manufacturer. Also, to have a more precise balancing process we can decrease the balancing voltage to 10 mV, but with the disadvantage of increasing the balancing time.

IV. CONCLUSIONS

For a fast balancing of four serially connected LiFePO₄ cells, the passive BMS proposed in this paper contains power resistors and power transistors as main components to dissipate the surplus of energy. The presented experimental results for different discharging / charging tests demonstrate that the

proposed BMS can maintain the cells voltages according to the nominal values from the datasheet.

The main advantage of the proposed BMS is that the balancing time is decreased by several times than commercially available BMS, although for large capacity cells like the ones which we have tested. This BMS can be used in maintenance operations for fast balancing if the BMS originally installed on a battery pack cannot do the balancing in a required time. Also, the BMS was designed to implement and analyze the performances of different cells balancing algorithms and estimation techniques for SOC, SOH, RUL, etc.

The major disadvantage of passive balancing is that the excessive energy is transformed into lot of heat by the power resistors and transistors, so the efficiency of the equalization process is low. This can be overcome by using active balancing of the cells, but with an increased complexity and costs of the balancing circuits and DC-DC converters.

ACKNOWLEDGMENT

This work has been partially funded by the European Regional Development Fund and the Romanian Government, through the Sectoral Operational Programme Increase of Economic Competitiveness (POSCCE), project “Research Centre for Advanced Materials, Products and Innovative Processes (CAMPUS)”, contract number 425/1/18.12.2012 and by University Politehnica of Bucharest, through the Excellence Research Grants Program (UPB-GEX-2016), project “Monitoring system through CAN-bus for parameters of an electric vehicle (CAN-EV)”, contract number 93/26.09.2016.

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